P-P-546D
June 13, 1975
SUPERSEDING
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August 13, 1963

FEDERAL SPECIFICATION

POLISH, AUTOMOBILE, LIQUID AND PASTE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types and two classes of polish suitable for cleaning, polishing, and protecting automobile finishes.

1.2 Classification. The automobile polish covered by this specification shall be of the following types and classes as specified (see 6.1).

Type I - Liquid polish.
Type II - Paste polish.

Class 1 - Without silicone.
Class 2 - With silicone.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

RR-S-366 - Sieves, Test.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-96 - Cans, Metal; 28 Gauge and Lighter.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods, Inspection, Sampling, and Testing: Method 6152, Accelerated Weathering (Enclosed Arc Apparatus).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

FSC 7930

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Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials:

D-523 - Specular Gloss.
D-1288 - Total Ash and Silica in Water Emulsion Waxes.
D-2834 - Nonvolatile Matter (Total Solids) in Water Emulsion Waxes, Solvent-Based Waxes and Polymer-Emulsion Floor Waxes.
E-70 - Determination of pH of Aqueous Solution with the Glass Electrode.
G-26 - Operating Light- and Water-Exposure Apparatus (Xenon-Arc Type) For Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Composition. The polishes shall consist of natural or synthetic waxes, or a mixture of both, volatile aliphatic solvents, polishing oils or silicone oils as applicable and mild abrasives. The type I polish shall be emulsified in water to form a stable, free-flowing emulsion that can be readily dispersed upon shaking.

3.2 Chemical and physical requirements. The polishes shall conform to the requirements specified in table I when tested as specified in table III.

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TABLE I. Chemical and physical requirements

Property	Type I				Type II			
	Class 1		Class 2		Class 1		Class 2	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Matter nonvolatile at 105°C., percent	15.0	40.0	17.0	45.0	25.0	45.0	27.0	50.0
Abrasive content, percent	5.0	15.0	7.0	16.0	5.0	15.0	7.0	17.0
Matter nonvolatile less abrasives (waxes, oils, and emulsifiers)	5.0	----	5.0	----	10.0	----	10.0	----
Silicone oil, percent	----	----	2.0	----	----	----	4.0	----
Fineness of abrasive material passing through U.S. Standard 170 sieve, percent	100.0	----	100.0	----	100.0	----	100.0	----
Material passing 270 sieve, percent	99.0	----	99.0	----	99.0	----	99.0	----
pH value	5.0	9.5	5.0	9.5	5.0	9.5	5.0	9.5

3.3 Performance requirements.

3.3.1 Gloss. The polish shall be equal or superior to the standard sample (see 6.3) when tested as specified in 4.4.3.

3.3.2 Smear resistance. When tested as specified in 4.4.3.2, the panels shall show no streaking or smearing.

3.3.3 Weatherability. The polish shall be equal or superior to the standard sample (see 6.3) when tested as specified in 4.4.4.

3.3.4 Cleaning ability. When tested as specified in 4.4.5, the panels shall show complete removal of the test soil.

3.3.5 Stability. When tested as specified in 4.4.6, the polish shall not show any difference when compared with untested polish.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for acceptance. From each inspection lot offered for Government inspection, four containers of one type shall be randomly selected, marked, and forwarded to the laboratory designated to perform acceptance testing.

4.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be scored as specified in table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an (AQL) of 4.0, expressed in terms of percent defective.

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4.3.1 Examination of closure, waterproofing and linebanding of shipping container. When shipping containers are required to comply with PPP-B-636, examination for defects in closure shall be in accordance with the appendix of that specification.

TABLE II. Examination of preparation for delivery

Examine	Defect
Marking	Incorrect; incomplete; illegible; omitted; improper size, location, sequence or method of application.
Containers	Components missing or damaged; leaking; dents or otherwise defective.
Materials	Components missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.
Contents	Not as specified.

4.4 Inspection of the end item.

4.4.1 Chemical and physical properties. The chemical and physical properties of the polishes outlined in table I will be tested as specified in table III.

TABLE III. Instructions for testing end item

Characteristic	Specification reference		Requirements applicable to		Number determinations per composite	Results reported as Pass or fail	Results reported as Numerically to nearest
	Requirement	Test method	Individ. unit	Lot aver.			
Matter nonvolatile at 105° C.	Table I	ASTM D-2834			2	--	0.1 percent
Abrasive content 1/	Table I	ASTM D-1288			2	--	0.1 percent
Matter nonvolatile less abrasive content (wax, oils, and emulsifiers)	Table I	2/			2	--	0.1 percent
Fineness of abrasive:	Table I	4.4.2					
Passing through U.S. Standard 170 sieve					2		0.1 percent
Passing through U.S. Standard 270 sieve					2		0.1 percent
pH value 3/	Table I	ASTM E-70			2		0.1 pH

1/ Abrasive content shall be taken as equal to total ash and silica.
2/ This value is obtained by subtracting the value obtained for abrasive content from the value obtained for matter nonvolatile at 105° C.
3/ The pH shall be taken on the undiluted polish for type I, and for type II the following procedure will be used: Five grams of the polish shall be added, with constant stirring to 100 ml. of freshly boiled distilled water in a beaker. Raise the temperature until wax is melted, stirring constantly for 10 minutes. Cool the wax mixture to 25 degrees C. plus or minus 5 degrees, decant the water from the cake of solidified wax and measure its pH with a glass electrode.

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4.4.2 Sieve test. For type I polish, weigh approximately 40 g into an 800 ml. beaker. Dilute polish to 500 ml. with hot, distilled water and digest on a steam bath for one hour. For type II polish, dissolve the 40 g of polish in 50 ml of kerosene and then disperse into 500 ml of kerosene. Vigorously stir to disperse solid matter and pour the slurry through a U.S. Standard 170 sieve conforming to RR-S-366. Wash with hot distilled water for type I or kerosene for type II. Determine visually if all solid matter passes through the U.S. Standard 170 sieve. Repeat this procedure with another sample of polish using a U.S. Standard 270 sieve. Wash material retained on the U.S. Standard 270 sieve with the appropriate solvent until no more solids pass through. Dry sieve contents at 105° C. for two hours, cool and transfer solid material into a tared container. Weigh and calculate percent passing through the U.S. Standard 270 sieve as follows:

$$\text{Percent passing through U.S. Standard 270 sieve} = \frac{\text{Sample weight-weight retained on 270 sieve} \times 100}{\text{Sample weight}}$$

4.4.3 Gloss.

4.4.3.1 Preparation of test panels. Obtain steel panels covered with black automobile 1/ lacquer and buff with 00 steel wool until a uniform matte finish is obtained. Wash with alcohol, rinse with tap water, and blot dry with a paper towel.

4.4.3.2 Application of polish. Using a soft cloth, apply a small amount of polish to the test panel, rubbing in a circular motion with the end of one finger. Use moderate pressure for a few seconds to apply a thin, even layer of polish. Allow the polished panel to dry for 5 to 6 minutes at room temperature and examine visually. Panel should be free from streaking and irregular deposits of polish. Buff lightly with lambs wool for 3 to 4 seconds and measure gloss. Prepare five duplicate panels of the standard sample (see 6.3) and the test polish.

4.4.3.3 Gloss measurement. Gloss measurement shall be carried out with an instrument conforming to ASTM D-523, using an angle of reflectance of 60 degrees. The polish under test shall produce a gloss equal or superior to that produced by standard sample (see 6.3).

4.4.4 Weatherability.

4.4.4.1 Weathering apparatus. Use a weathering apparatus for exposing polished panels to cycles of carbon arc light and water spray as specified in Fed. Test Method Std. No. 141A, Method 6152. Alternately, the same exposure may be produced with a xenon-arc apparatus conforming to ASTM G-26.

4.4.4.2 Gloss retention after weathering. Prepare five duplicate panels of the standard sample (see 6.3) and the test polish and place in the weathering apparatus. Subject them to a cycle of 102 minutes of carbon arc light and 18 minutes of water spray for a 48-hour period. Remove the panels from the apparatus and wipe them lightly with a soft cloth. Let the panels dry and then determine the gloss of each panel within 24 hours as specified in 4.4.3.3. The polish under test shall retain gloss equal or superior to that of the standard sample.

4.4.5 Cleaning ability.

4.4.5.1 Preparation of test panels. Prepare white panels as described in 4.4.3.1, except that buffing to a matte finish is not required; apply drops of a 1 percent solution of CSMA standard soil 2/ in tap water over the surface of the panel and allow to dry at room temperature. Then allow to stand for 16 hours in an oven maintained at 105° C. The dried, baked soil should appear as a pattern of dirty spots on the test panel.

1/ Such 4 x 6" panels are available from the Ford Motor Company, Central Labs Services, 3001 Miller Road, Dearborn, MI 48121.

2/ Available from the Chemical Specialties Manufacturers Association, 1001 Connecticut Ave., NW, Washington, DC 20036. Note also that is a suspension rather than a true solution and should be shaken before application.

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4.4.5.2 Application of polish. Apply polish as described in 4.4.3.2. After buffing for 3 to 4 seconds with lambs wool, examine the panel visually for traces of unremoved soil. The panel should be completely clean.

4.4.6 Stability.

4.4.6.1 Freeze-thaw cycle. Place an unopened container of auto polish in a freezer maintained at -5 to -8° C. and allow to stand for 16 hours. Remove from the freezer and allow to stand at room temperature for eight hours. Repeat the cycle three times.

4.4.6.2 Application. Apply as described in 4.4.3.1 the polish subjected to the freeze-thaw cycle and compare it with unfrozen polish similarly applied for ease of application, streaking, smearing, and gloss.

4.4.6.3 Visual inspection. Examine the frozen and unfrozen type I polish for separation, or if layering exists normally, for ease of remixing upon shaking. Examine the type II polish for separation of liquid.

4.4.6.4 Heat stability. Place 100 to 150 ml of polish in closed container and allow to stand in an oven maintained at 50° C. (122° F.) for 48 hours. Remove and allow to cool, then examine for evidence of layering, separation or discoloring.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I. The liquid polish shall be furnished in 1-pint, 1-quart or 1-gallon (see 6.2) oblong cans conforming to type V, class 4, with inner seal and exterior coating plan B of PPP-C-96, except that 0.25 pound electrolytic tinplate, terneplate, or blackplate shall not be used. The 1-pint cans of polish shall be packaged in accordance with the interior packaging requirements of the Appendix of PPP-C-96.

5.1.1.2 Type II. The paste polish shall be furnished in 8-ounce or 1-pound (see 6.2) round cans conforming to type V, class 1 or 2, exterior coating plan B of PPP-C-96, except that 0.25 pound electrolytic tinplate, terneplate, or blackplate shall not be used. The 8-ounce cans of polish shall be packaged in accordance with the interior packaging requirements of the Appendix of PPP-C-96.

5.1.2 Level B. The paste polish shall be in accordance with paragraphs 5.1.1.1 and 5.1.1.2 except that exterior coating plan A of PPP-C-96 shall be specified.

5.1.3 Level C. Automobile polish of one type and class shall be furnished in containers to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Automobile polish of one type only, packaged as specified in 5.1, shall be packed in accordance with the level A requirements in the Appendix of PPP-C-96.

5.2.2 Level B. Automobile polish of one type only, packaged as specified in 5.1, shall be packed in accordance with the level 3 requirements in the Appendix of PPP-C-96.

5.2.3 Level C. The polish in quantities as specified (see 6.2) packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

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5.3 Marking. In addition to any special marking required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as specified (see 6.2).

6. NOTES

6.1 Intended use. The automobile cleaner polishes covered by this specification are intended for cleaning, polishing and protecting the finish of automobiles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Packing and packaging required (see section 5).
(d) Quantities required.
(e) Marking document required (see 5.3).

6.3 Standard sample. One pound quantities of the standard sample are available at cost from the Federal Supply Service, General Services Administration, Washington, DC 20406. The same standard sample, a type II, class 2 material is furnished for all types and classes.

MILITARY INTEREST:

The Defense Department has waived coordination on amendments or revisions of this document until further notice.

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

GSA-FSS
VA-DMS
DOT-MVP

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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